



DEEPX DX-M1 Quick User Manual for Raspberry Pi 5

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DEEPX.ai

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1. Introduction

The **DEEPX DX-M1** is a middle-entry edge NPU designed for high-performance AI acceleration. This manual provides technical guidance for integrating the **DEEPX DX-M1** Accelerator Card with the Raspberry Pi 5 via PCIe M.2. It covers hardware architecture, driver/firmware installation, and the software setup required for AI inference, including sample applications and real-time monitoring via `dxtop`.

2. System Overview

This chapter provides a technical overview of the DEEPX DX-M1 accelerator and its integration environment.

2.1 Hardware Architecture: DX-M1 M.2

The system utilizes a high-performance heterogeneous architecture specifically engineered to support demanding edge AI workloads.

2.1.1 DX-M1 M.2 Accelerator Card Specifications

The DX-M1 is a high-efficiency AI inference accelerator designed for seamless integration into edge computing environments through its optimized internal architecture.

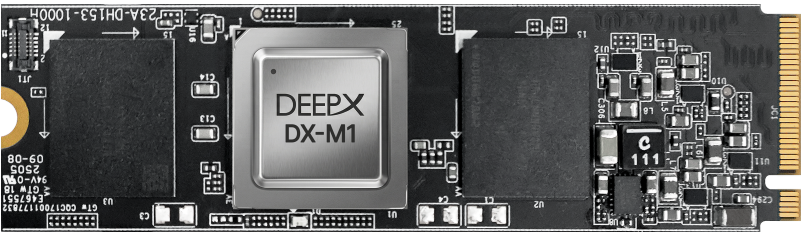


Figure. DX-M1 M.2 Architecture

Table. DX-M1 M.2 Technical Specification

Features	Details
AI Performance	25 TOPS (INT8)
Host Interface	PCIe Gen3 x4 (Supports Gen 1/2/3 & x1/x2/x4)
Memory	4GB LPDDR5 (5600 MT/s)
Power Consumption	1W (min) ~ 5W (max)
Form Factor	M.2 2280 (M Key), 22 x 80mm
OS Support	Windows 11/10 Debian-based Linux (Ubuntu 24.04/22.04/20.04 LTS) Yocto Project

Features	Details
AI Frameworks	Ultralytics, PyTorch, TensorFlow, ONNX, Keras
System Support	x86 and ARM-based Architectures

2.2 Hardware Installation and I/O Setup

This section outlines the physical integration of the DEEPX DX-M1 accelerator. Follow these steps to ensure the host system and hardware components are properly configured for a stable edge AI environment.

2.2.1 Step-by-Step Connection Guide

Step 1. Raspberry Pi 5 Initial Setup

Complete the basic OS and power configuration by referring to the official documentation:

- Getting started with your Raspberry Pi (<https://www.raspberrypi.com/documentation/computers/getting-started.html>)

Step 2. Connecting the DX-M1 M.2 Module

Once the host is prepared, install the DX-M1 M.2 NPU module via the PCIe M.2 HAT as shown below.

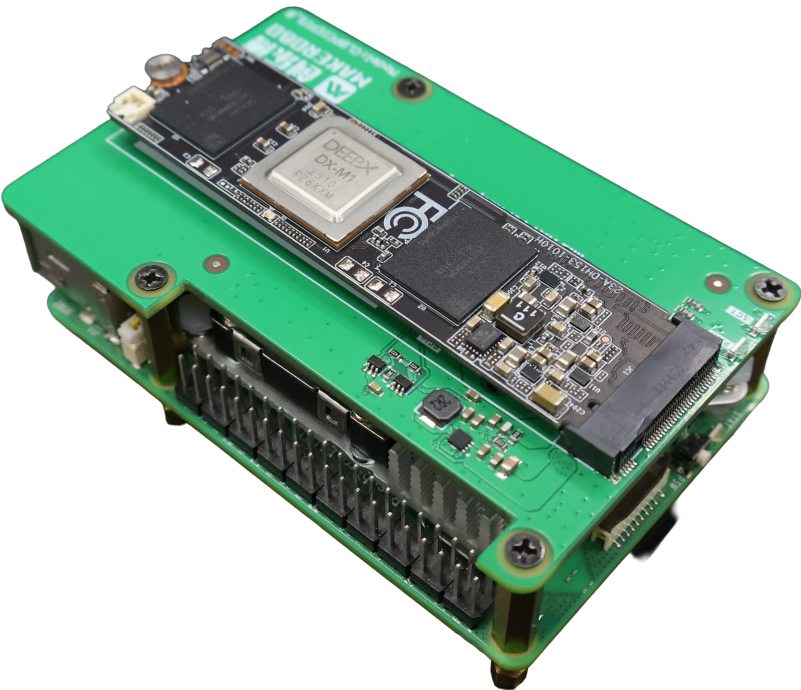


Figure. DEEPX DX-M1 AI Accelerator for Raspberry Pi 5

3. Getting Started

This chapter provides a concise guide to configuring the DEEPX DX-M1 development environment. It covers the essential steps from verifying system requirements to executing your first AI inference benchmarks. It assumes the DX-M1 M.2 module has already been physically installed as described in [System Overview](#).

3.1 System Prerequisites

Before proceeding with the driver installation, ensure the host system is properly configured by following these environment preparation steps.

3.1.1 Environment Preparation

Step 1. Install Raspberry Pi OS

- If you have not already done so in [System Overview](#), download and install the OS using the **Raspberry Pi Imager**.
- Link: <https://www.raspberrypi.com/software/>

Step 2. Connect to the Internet

- Verify that the Raspberry Pi has an active **Internet connection**.
 - A stable connection is required to access external repositories and download necessary dependencies during the setup process.
-

3.2 Software Setup and Demo Execution

This section provides a brief guide to installing the necessary drivers and executing AI inference benchmarks using the DX-M1 accelerator.

3.2.1 Install DEEPX M1 Driver and Runtime

Follow these steps to configure the official repository and install the **DEEPX Runtime (DXRT)** library, known as libdxrt, along with the necessary driver modules.

Step 1. Download the debian package

Execute the following commands to download latest debian package.

```
# Download debian package to Download folder
$ wget -P ~/Downloads https://github.com/DEEPX-AI/dx_rt/raw/refs/heads/main/release/3.3.2/libdxrt_3.3.2_all.deb

$ wget -P ~/Downloads https://github.com/DEEPX-AI/dx_rt_npu_linux_driver/raw/refs/heads/main/release/2.4.0/dxrt-driver-dkms_2.4.0-2_all.deb
```

```
deepx@deepx:~$ wget -P ~/Downloads https://github.com/DEEPX-AI/dx_rt_npu_linux_driver/raw/refs/heads/main/release/2.4.0/dxrt-driver-dkms_2.4.0-2_all.deb
--2026-05-07 15:18:22-- https://github.com/DEEPX-AI/dx_rt_npu_linux_driver/raw/refs/heads/main/release/2.4.0/dxrt-driver-dkms_2.4.0-2_all.deb
Resolving github.com (github.com)... 64:ff9b:14c8:f5f7, 20.208.245.247
Connecting to github.com (github.com)[64:ff9b:14c8:f5f7]:443... connected.
HTTP request sent, awaiting response... 302 Found
location: https://raw.githubusercontent.com/DEEPX-AI/dx_rt_npu_linux_driver/refs/heads/main/release/2.4.0/dxrt-driver-dkms_2.4.0-2_all.deb [following]
--2026-05-07 15:18:22-- https://raw.githubusercontent.com/DEEPX-AI/dx_rt_npu_linux_driver/refs/heads/main/release/2.4.0/dxrt-driver-dkms_2.4.0-2_all.deb
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 2606:50c0:8003::154, 2606:50c0:8000::154, 2606:50c0:8001::154, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)[2606:50c0:8003::154]:443... connected.
HTTP request sent, awaiting response... 200 OK
length: 146436 (143K) [application/octet-stream]
Saving to: '/home/deepx/Downloads/dxrt-driver-dkms_2.4.0-2_all.deb'

dxrt-driver-dkms_2.4.0-2_all.deb      100%[=====] 143.08K  ---KB/s   in 0.1s
2026-05-07 15:18:28 (1.46 MB/s) - '/home/deepx/Downloads/dxrt-driver-dkms_2.4.0-2_all.deb' saved [146436/146436]

deepx@deepx:~$ wget -P ~/Downloads https://github.com/DEEPX-AI/dx_rt/raw/refs/heads/main/release/3.3.1/libdxrt_3.3.1_all.deb
--2026-05-07 15:18:54-- https://github.com/DEEPX-AI/dx_rt/raw/refs/heads/main/release/3.3.1/libdxrt_3.3.1_all.deb
Resolving github.com (github.com)... 64:ff9b:14c8:f5f7, 20.208.245.247
Connecting to github.com (github.com)[64:ff9b:14c8:f5f7]:443... connected.
HTTP request sent, awaiting response... 302 Found
location: https://raw.githubusercontent.com/DEEPX-AI/dx_rt/refs/heads/main/release/3.3.1/libdxrt_3.3.1_all.deb [following]
--2026-05-07 15:18:55-- https://raw.githubusercontent.com/DEEPX-AI/dx_rt/refs/heads/main/release/3.3.1/libdxrt_3.3.1_all.deb
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 2606:50c0:8003::154, 2606:50c0:8000::154, 2606:50c0:8001::154, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)[2606:50c0:8003::154]:443... connected.
HTTP request sent, awaiting response... 200 OK
length: 22016682 (21M) [application/octet-stream]
Saving to: '/home/deepx/Downloads/libdxrt_3.3.1_all.deb'

libdxrt_3.3.1_all.deb      100%[=====] 21.08M  8.43MB/s   in 2.5s
2026-05-07 15:18:59 (8.43 MB/s) - '/home/deepx/Downloads/libdxrt_3.3.1_all.deb' saved [22016682/22016682]
```

Figure. Download Debian Package

Step 2. Installation of DXRT and Driver

Install the DEEPX Runtime (DXRT) library and the NPU driver.

```
$ sudo apt install -y ~/Downloads/dxrt-driver-dkms_2.4.0-2_all.deb
$ sudo apt install -y ~/Downloads/libdxrt_3.3.2_all.deb
```

```

-- Set non-toolchain portion of runtime path of "/usr/local/bin/examples/run_async_model_bufcount" to "/usr/local/lib:/usr/share/libdrt/build_aarch64/lib"
-- Installing: /usr/share/libdrt/bin/examples/run_async_model_bufcount
-- Set non-toolchain portion of runtime path of "/usr/share/libdrt/bin/examples/run_async_model_bufcount" to "/usr/local/lib:/usr/share/libdrt/build_aarch64/lib"
-- Installing: /usr/local/bin/examples/run_batch_model
-- Set non-toolchain portion of runtime path of "/usr/local/bin/examples/run_batch_model" to "/usr/local/lib:/usr/share/libdrt/build_aarch64/lib"
-- Installing: /usr/share/libdrt/bin/examples/run_batch_model
-- Set non-toolchain portion of runtime path of "/usr/share/libdrt/bin/examples/run_batch_model" to "/usr/local/lib:/usr/share/libdrt/build_aarch64/lib"
-- Installing: /usr/local/bin/examples/run_sync_model
-- Set non-toolchain portion of runtime path of "/usr/local/bin/examples/run_sync_model" to "/usr/local/lib:/usr/share/libdrt/build_aarch64/lib"
-- Installing: /usr/share/libdrt/bin/examples/run_sync_model
-- Set non-toolchain portion of runtime path of "/usr/share/libdrt/bin/examples/run_sync_model" to "/usr/local/lib:/usr/share/libdrt/build_aarch64/lib"
-- Installing: /usr/local/bin/examples/run_sync_model_bound
-- Set non-toolchain portion of runtime path of "/usr/local/bin/examples/run_sync_model_bound" to "/usr/local/lib:/usr/share/libdrt/build_aarch64/lib"
-- Installing: /usr/share/libdrt/bin/examples/run_sync_model_bound
-- Set non-toolchain portion of runtime path of "/usr/share/libdrt/bin/examples/run_sync_model_bound" to "/usr/local/lib:/usr/share/libdrt/build_aarch64/lib"
-- Installing: /usr/local/bin/examples/run_sync_model_output
-- Set non-toolchain portion of runtime path of "/usr/local/bin/examples/run_sync_model_output" to "/usr/local/lib:/usr/share/libdrt/build_aarch64/lib"
-- Installing: /usr/share/libdrt/bin/examples/run_sync_model_output
-- Set non-toolchain portion of runtime path of "/usr/share/libdrt/bin/examples/run_sync_model_output" to "/usr/local/lib:/usr/share/libdrt/build_aarch64/lib"
-- Installing: /usr/local/bin/examples/run_sync_model_bufcount
-- Set non-toolchain portion of runtime path of "/usr/local/bin/examples/run_sync_model_bufcount" to "/usr/local/lib:/usr/share/libdrt/build_aarch64/lib"
-- Installing: /usr/share/libdrt/bin/examples/run_sync_model_bufcount
-- Set non-toolchain portion of runtime path of "/usr/share/libdrt/bin/examples/run_sync_model_bufcount" to "/usr/local/lib:/usr/share/libdrt/build_aarch64/lib"
-- Installing: /usr/local/bin/examples/run_sync_model_memory
-- Set non-toolchain portion of runtime path of "/usr/share/libdrt/bin/examples/run_sync_model_memory" to "/usr/local/lib:/usr/share/libdrt/build_aarch64/lib"
-- Installing: /usr/share/libdrt/bin/examples/run_sync_model_memory
-- Set non-toolchain portion of runtime path of "/usr/share/libdrt/bin/examples/run_sync_model_memory" to "/usr/local/lib:/usr/share/libdrt/build_aarch64/lib"
-- Installing: /usr/local/bin/examples/multi_input_model_inference
-- Set non-toolchain portion of runtime path of "/usr/local/bin/examples/multi_input_model_inference" to "/usr/local/lib:/usr/share/libdrt/build_aarch64/lib"
-- Installing: /usr/share/libdrt/bin/examples/multi_input_model_inference
-- Set non-toolchain portion of runtime path of "/usr/share/libdrt/bin/examples/multi_input_model_inference" to "/usr/local/lib:/usr/share/libdrt/build_aarch64/lib"
-- Installing: /usr/local/bin/examples/check_versions
-- Set non-toolchain portion of runtime path of "/usr/local/bin/examples/check_versions" to "/usr/local/lib:/usr/share/libdrt/build_aarch64/lib"
-- Installing: /usr/share/libdrt/bin/examples/check_versions
-- Set non-toolchain portion of runtime path of "/usr/share/libdrt/bin/examples/check_versions" to "/usr/local/lib:/usr/share/libdrt/build_aarch64/lib"
[INFO] Creating symlink to compile_commands.json
[INFO] [Install python package ...]
[INFO] Checking Python version: /usr/bin/python3
[INFO] Python version (3.13.5) meets the requirements.
dx-engine 3.3.1
[INFO] The Python package has been installed in the following Python version folder.
: pip 25.1.1 from /usr/lib/python3/dist-packages/pip (python 3.13)
[HINT] To install it in a Python virtual environment,
[HINT] you need to activate the virtual environment and then
[HINT] reinstall it by running 'python3 -m pip install .' from the python_package folder.
[SUCCESS] [Installed python package]
[INFO] Service mode: on
[SUCCESS] [Created dxrt.service file]
[SUCCESS] [Reloaded systemd daemon]
[INFO] [dxrt service not enabled]
Created symlink '/etc/systemd/system/multi-user.target.wants/dxrt.service' -> '/etc/systemd/system/dxrt.service'.
[SUCCESS] [Enabled the dxrt service (dxrt0)]
[INFO] [dxrt service not started]
[SUCCESS] [Started the dxrt service (dxrt0)]
[SUCCESS] Build mode completed
Notice: Download is performed unsandboxed as root as file '/home/deepx/Downloads/libdrt_3.3.1_all.deb' couldn't be accessed by user '_apt'. - pkgAcquire::Run (13: Permission denied)
deepx@deepx:~$

```

Figure. Runtime Library ('libdrt') Installation Progress

Step 3. Hardware Status Verification

Confirm that the DX-M1 module is correctly interfaced and recognized by the system.

```
$ dxrt-cli -s
```

```

deepx@deepx:~ $ dxrt-cli -s
DXRT v3.3.1
=====
* Device 0: M1, Accelerator type
----- Version -----
* RT Driver version   : v2.4.0
* PCIe Driver version : v2.2.0
-----
* FW version          : v2.5.6
----- Device Info -----
* Memory : LPDDR5x 6000 Mbps, 3.92GiB
* Board  : M.2, Rev 1.0
* Chip Offset : 0
* PCIe    : Gen2 X1 [01:00:00]

NPU 0: voltage 750 mV, clock 1000 MHz, temperature 33'C
NPU 1: voltage 750 mV, clock 1000 MHz, temperature 33'C
NPU 2: voltage 750 mV, clock 1000 MHz, temperature 33'C
=====
deepx@deepx:~ $ 

```

Figure. Hardware Verification using `dxrt-cli -s` Command

3.2.2 Optional: Firmware Update

To ensure full compatibility between the `libdxrt` library and the hardware, the device firmware must meet the minimum version requirements.

Prerequisites for Update

Perform a firmware update if any of the following conditions occur.

- **Version Mismatch:** The current firmware version is lower than the minimum requirement.
- **Execution Error:** A `[dxrt-exception] Invalid Operation` error occurs during model execution.

Firmware Download

The latest firmware binaries and guides are available at the official DEEPX GitHub repository.

- **URL:** https://github.com/DEEPX-AI/dx_fw

Quick Update Procedure

After downloading the firmware, execute the following commands.

```
# Step 1. Stop service
$ systemctl stop dxrt.service

# Step 2. Update firmware
$ dxrt-cli -u <path_to_fw_bin>

# Step 3. Restart service
$ systemctl start dxrt.service
```

CAUTION

Do not power off the device during the update to prevent permanent hardware damage.

3.2.3 Execution of Sample Applications

AI Inference Demo (`dx_app`)

The `dx_app` demonstrates real-time AI inference capabilities. The execution script automatically handles the acquisition of sample video assets and pre-compiled models.

```
$ mkdir ~/sample
$ cd ~/sample
$ git clone --recurse-submodules https://github.com/DEEPX-AI/dx_app.git
$ cd dx_app
$ ./install.sh --all
$ ./run_demo.sh
```

```

12: Super Resolution      (ESPCN-X4)
13: Image Enhancement    (Zero-DCE)

[ Recognition ]
14: Embedding            (ArcFace)
15: Attribute Recognition (DeepMAR-ResNet50)
16: Person Re-ID         (CasViT-T)

[ PPU ]
17: PPU Pipeline         (YOLOv7-PPU)

Select task [0-17, default: 0]:
[INFO] Task: Object Detection      (YOLOv7)

-----
[ Stage 2 / 3 ] Select Execution Mode
Task : Object Detection      (YOLOv7)
-----

1: cpp_sync
2: cpp_async
3: py_sync
4: py_async
5: py_sync_cpp_postprocess
6: py_async_cpp_postprocess

Select mode [1-6, default: 1]:
[INFO] Mode: cpp_sync

-----
[ Stage 3 / 3 ] Select Input Type
Task : Object Detection      (YOLOv7)
Mode : cpp_sync
-----

1: video (assets/videos/snowboard.mp4)
2: image (sample/img/sample_street.jpg)

Select input [1=video, 2=image, default: 1]:
[INFO] Input: video (assets/videos/snowboard.mp4)

Enable verbose log output? [y/N, default: N]:

-----
Task : Object Detection      (YOLOv7)
Mode : cpp_sync
Input : video (assets/videos/snowboard.mp4)
Cmd   : /home/deepx/dev/dx_app/bin/yolov7_sync -m assets/models/YoloV7.dnn -v assets/videos/snowboard.mp4

-----
[INFO] Model loaded: assets/models/YoloV7.dnn
[INFO] Model input size (WxH): 640x640

[INFO] Starting inference...

(yolov7_sync:23211): GLib-GObject-CRITICAL **: 17:27:35.314: g_object_unref: assertion 'G_IS_OBJECT (object)' failed

```

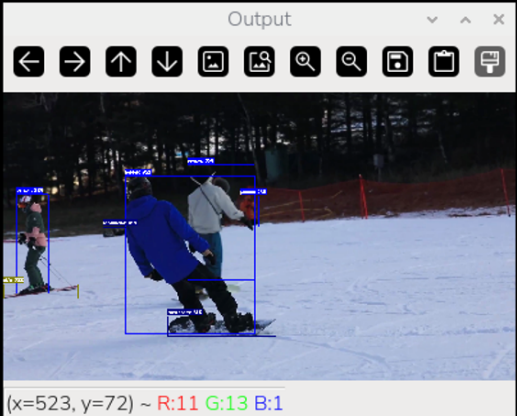


Figure. `dx\_app` Demo Output: Terminal Performance Summary (top) and Real-time Object Detection Result (bottom)

GStreamer Plugin Demo (dx_stream)

The `dx_stream` sample is optimized for high-performance video streaming pipelines utilizing the GStreamer framework.

```

$ cd ~/sample
$ git clone --recurse-submodules https://github.com/DEEPX-AI/dx_stream.git
$ cd dx_stream
$ ./install.sh

```



```
$ ./build.sh
$ ./run_demo.sh
```

```

deepx@deepx: ~/sample/dx_stream
File Edit Tabs Help
deepx@deepx:~/sample/dx_stream $ ./run_demo.sh
~/sample/dx_stream ~/sample/dx_stream
[INFO] DX_STREAM_PATH: /home/deepx/sample/dx_stream
[INFO] Models and Videos directory already exists. Skipping download.
0: Object Detection (YOLOv5s)
1: Object Detection (YOLOv5s with PPU)
2: Face Detection (YOLOV5S_Face)
3: Face Detection (SCRFD500M with PPU)
4: Pose Estimation (YOLOV5Pose)
5: Pose Estimation (YOLOV5Pose with PPU)
6: Multi-Object Tracking
7: Semantic Segmentation
8: Multi-Channel Object Detection
9: Multi-Channel RTSP
-: secondary mode
which AI demo do you want to run:(timeout:10s, default:0)

```



Figure. `dx\_stream` GStreamer Pipeline: Plugin menu selection (top) and multi-object detection result (bottom)

3.2.4 System Monitoring

NPU Utilization (`dxtop`)

The `dxtop` utility provides a real-time CLI (Command Line Interface) to monitor NPU utilization, temperature, and operational status.

```
$ dxtop
```

```
[DX-TOP] (q) Quit (n) Next Page (p) Prev Page (h) Help
Thu May 7 15:26:57 2026
DX-RT: v3.3.1 NPU Device driver: v2.4.0 DX-TOP: v1.1.0
-----
Device :0 Variant: M1 PCIe Bus Number: 01:00:00 Firmware: v2.5.6
NPU Memory: [ 4.00 KiB / 3.92 GiB (0.00%) ]
Core :0 Util: 0.0% Temp: 33 °C Voltage: 750 mV Clock: 1000 MHz
Core :1 Util: 0.0% Temp: 33 °C Voltage: 750 mV Clock: 1000 MHz
Core :2 Util: 0.0% Temp: 33 °C Voltage: 750 mV Clock: 1000 MHz
-----
```

Figure. Real-time NPU Resource Monitoring via ``dxtop``

3.2.5 PCIe Gen3 Configuration

To maximize the throughput of the DX-M1 accelerator, it is recommended to manually force the interface to **PCIe Gen3**.

Manual Configuration via `config.txt`

This method involves editing the system configuration file to enforce Gen3 speeds manually.

Step 1. Open the configuration file

```
$ sudo nano /boot/firmware/config.txt
```

NOTE

In older versions of the OS, the path may be `/boot/config.txt`

Step 2. Add or modify the PCIe parameters

```
# Enable the PCIe external connector (M.2 HAT)
dtparam=pciex1
```

```
# Force PCIe Gen3 speed (Required for 25 TOPS performance)
dtparam=pciex1_gen=3
```

Step 3. Save and Exit

Press **Ctrl + O**, then **Enter** to save the changes, and Press **Ctrl + X** to exit the editor.

Step 4. Reboot the system

```
$ sudo reboot
```

Verification of PCIe Link Speed

After rebooting, you can verify if the PCIe Gen3 configuration has been applied correctly by checking the Link Status.

```
$ lspci -vv | grep -i LnkSta:
```

- If the output shows **Speed 8GT/s**, the system is successfully operating at **PCIe Gen3** speeds.
- If the output shows **Speed 5GT/s**, it is operating at **PCIe Gen2**.

NOTE

To achieve the maximum AI performance of 25 TOPS, the link speed **must** be confirmed as **8GT/s**.

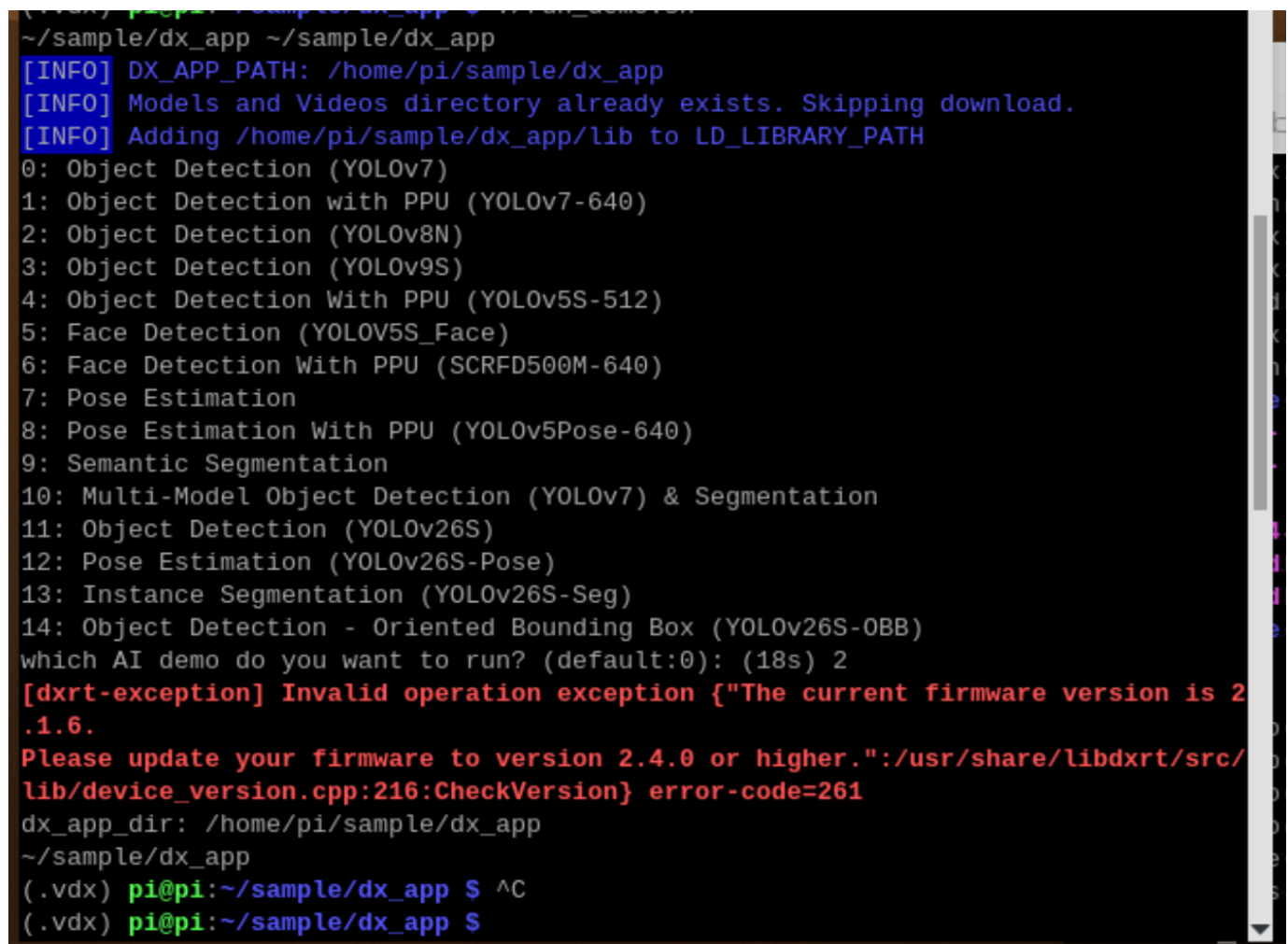
If you run into errors during setup, see [Appendix. Troubleshooting](#).

4. Appendix. Troubleshooting

4.1 Q1. Firmware Version Mismatch Error Message

Issue Description

When executing the AI demo application, the program terminates with a red `[dxrt-exception]` error message. This occurs when the current hardware firmware version is incompatible with the runtime library (`libdxrt`) requirements.



```
~/sample/dx_app ~/sample/dx_app
[INFO] DX_APP_PATH: /home/pi/sample/dx_app
[INFO] Models and Videos directory already exists. Skipping download.
[INFO] Adding /home/pi/sample/dx_app/lib to LD_LIBRARY_PATH
0: Object Detection (YOLOv7)
1: Object Detection with PPU (YOLOv7-640)
2: Object Detection (YOLOv8N)
3: Object Detection (YOLOv9S)
4: Object Detection With PPU (YOLOv5S-512)
5: Face Detection (YOLOv5S_Face)
6: Face Detection With PPU (SCRFD500M-640)
7: Pose Estimation
8: Pose Estimation With PPU (YOLOv5Pose-640)
9: Semantic Segmentation
10: Multi-Model Object Detection (YOLOv7) & Segmentation
11: Object Detection (YOLOv26S)
12: Pose Estimation (YOLOv26S-Pose)
13: Instance Segmentation (YOLOv26S-Seg)
14: Object Detection - Oriented Bounding Box (YOLOv26S-0BB)
which AI demo do you want to run? (default:0): (18s) 2
[dxrt-exception] Invalid operation exception {"The current firmware version is 2.1.6. Please update your firmware to version 2.4.0 or higher." : /usr/share/libdxrt/src/lib/device_version.cpp:216:CheckVersion} error-code=261
dx_app_dir: /home/pi/sample/dx_app
~/sample/dx_app
(.vdx) pi@pi:~/sample/dx_app $ ^C
(.vdx) pi@pi:~/sample/dx_app $
```

Figure. Firmware Version Mismatch Error in Demo Execution

Cause

- **Version Conflict:** The system has detected that the installed firmware is lower than the minimum required version needed.

Solution

- To resolve this, you must perform a manual firmware upgrade to ensure the hardware can communicate correctly with the latest SDK features. Refer to **Section. Optional: Firmware Update**.
 - If the issue persists after the firmware update, contact DEEPX Technical Support at tech-support@deepx.ai.
-